

NOVA 200

Automatic Section Control Computer

Installation & User Manual

Table of Contents

1. Symbols & Abbreviations 05
   1.1. Symbols Used 05

2. Before Using the Product 06
   2.1 Important safety instructions 06
      2.1.1 Risk of fire 06
      2.1.2 Electrostatic Discharge 06
      2.1.3 Usage/Installation Pre-requisites 06

3. About the Product 07
   3.1 About NOVA 200 (Automatic Section Control Computer) 07
   3.2 Product Overview 08

4. Installation 10
   4.1 Device Installation 10

5. Software Operations 12
   5.1 Continue Job 12
   5.2 Start New Job 12
   5.3 Data Display 12
      5.3.1 GPS Data 13
      5.3.2 Job Data 13
      5.3.3 Device Data 13
      5.3.4 Field Overview 13
   5.4 Tank Settings 13
      5.4.1 Refill Tank 13
      5.4.2 Set Tank Level 13
   5.5 Job Menu 14
      5.5.1 Select Job 14
      5.5.2 Edit Job 14
      5.5.3 Select Memory 14
      5.5.4 Boundry Menu 14

Table of Contents

5.6 Settings 15

5.6.1 Tractor 15

5.6.2 Rate Control Settings 16

5.6.3 Implement 18

5.6.4 User Settings 22

5.6.5 Section Control Settings 24

5.6.6 Advanced Settings 28

6. Operating Instructions 32

6.1 First Operation 32

6.1.1 Power On 32

6.1.2 Tractor Geometry Settings 32

6.1.3 Implement Geometry&Configuration 33

6.1.4 Rate Control Settings 33

6.1.5 Electrovalve Tests 34

6.1.6 Valve ON/OFF Delay Measurement 34

6.1.7 Aggressiveness Fine Adjustment 35

6.1.8 Section Control Parameters 35

6.1.9 Screen Configuration Tank Level 36

6.1.10 Tank Settings 36

6.1.11 Select Tractor&Implement 37

6.1.12 Select Application Rate 37

6.1.13 Start New Job 38

6.1.14 First Operation – Sprayer Critical Check (Quick List) 38

6.2 Operating Features 39

6.2.1 Boundary Creation 34

6.1.2 Boundary Selection 34

6.2.3 Automatic Section Control 40

6.2.4 Automatic Rate Control 40

6.2.5 A-B Line 41

6.2.6 Pause Point 42

6.2.7 Tank Fill-Refill 42

7. Package Contents 43

8. Disposal of Used Packaging Material 45

About this Document

Document Information

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2.			
3.			
4.			

1. Symbols & Abbreviations

This manual is for qualified technicians only. The tasks described in this manual may only be performed by qualified technicians.

1.1 Symbols Used

The following types of safety instructions and general information appear in this document described as below:

Symbol	Type of Hazard	Description
	**DANGER**	- Indicates a hazardous situation that needs immediate attention. - If not avoided, serious injury or even death may occur.
	**WARNING**	- Indicates a hazardous situation that could lead to death or serious injury if precautions are not taken.
	**CAUTION**	- Indicates a hazardous situation that might result in minor or moderate injury if not avoided.
	**NOTE**	- Provides you tips that are valuable for the optimal operation of your product.

2. Before Using the Product

	Carefully read all the safety instructions. Otherwise, there exists a risk of an accident or a fatal injury.
	Never clean equipment with direct water jets. This may cause irreparable damage to the product.
	Do not use solvents or fuel to clean the case outer surface. It may cause serious damage to the product assembly.
	Comply with the specified power voltage (12 VDC).
	Only use Tim spare parts.

2.1. Important safety instructions

2.1.1 Risk of fire

	*This is an electrical product and is susceptible to fires.*

- Do not store or place any flammable substances near the product components or wiring.
- Keep cigarettes and other open flames away from the product. There is a danger of fire damaging the product or wiring.
- Do not expose the product to temperatures in excess of 70°C.
- Reverse polarity will cause irreversible damage to the circuitry.
- Do not store or operate in a location where it may be at risk of getting in contact wit chemicals

2.1.2 Electrostatic Discharge

	*This is an electrical product and is susceptible to static charge buildup.*

- Electronic components on the Printed Circuit Boards within the product are susceptible to damage from Electrostatic Discharge (ESD).
- Observe ESD precautions when handling ESD sensitive components. Refer to site specific procedures if applicable.

2.1.3 Usage/Installation Pre-requisites

	*This is an electrical product and is susceptible to static charge buildup.*

Before Usage/installation, carefully verify the below-mentioned prerequisites.

1. There is no excessive humidity in the usage/installation area.
2. There are no flammables or explosives stored nearby.
3. There is no excessive dust and dirt in the area.
4. All federal, state and local site safety requirements are in place.

About NOVA 200 (Automatic Section Control Computer)

Nova 200's automatic section control stops spraying automatically in overlapping areas, preventing resource waste and providing environmentally sensitive application. Thanks to GNSS-supported features, it provides precise location tracking and field variability mapping capabilities. This enables variable rate applications to different zones, optimizing input usage. With the capability to control up to 7 sections, precise rate adjustments, and real-time data processing, Nova 200 transforms your spraying machines into smart agricultural tools. Its user-friendly menu, multi-sensor support, and robust electronic infrastructure offer a powerful solution in the digital transformation of modern farming.

Key capabilities include:

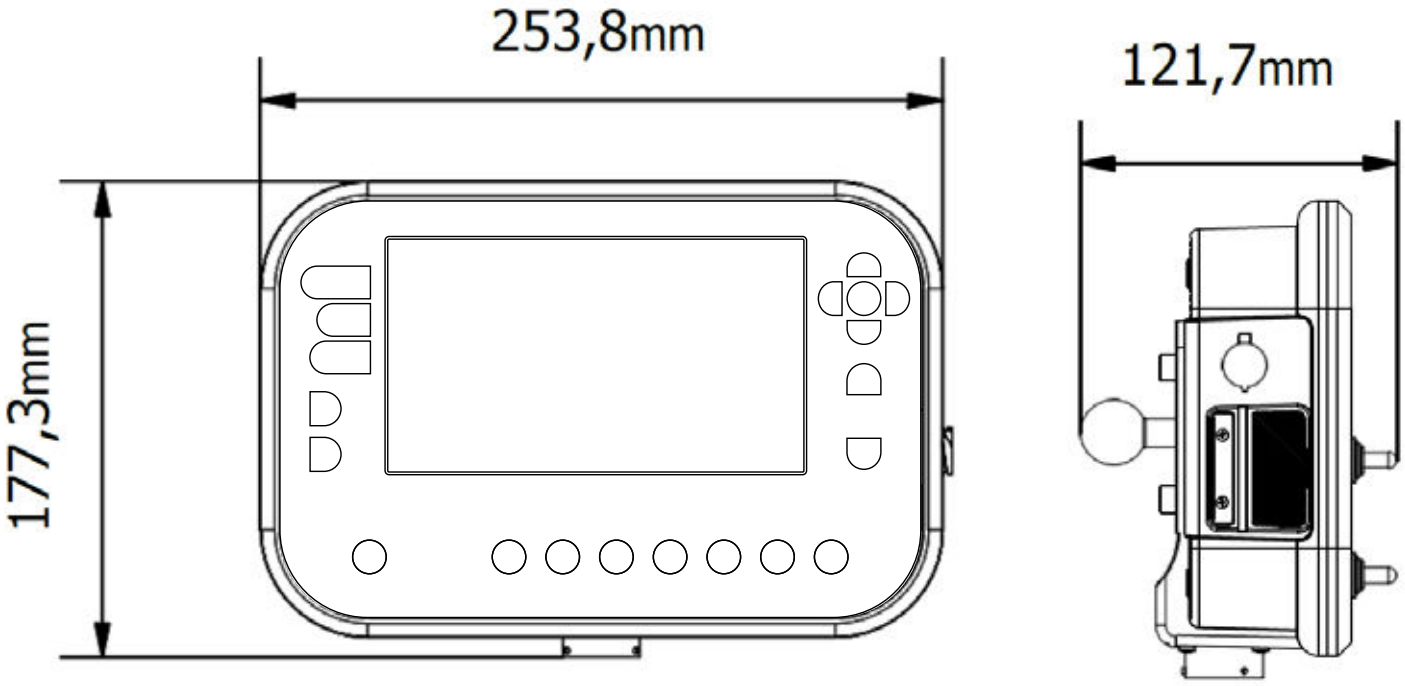
- **Automatic Section Control:** Eliminate overlap and input waste with intelligent section automation that stops spraying instantly in already covered zones — ensuring efficient and eco-friendly operations.
- **GNSS-Based Mapping & Location Tracking:** Harness high-precision GNSS technology to track field position in real time, map field variability, and enhance overall spraying accuracy.
- **Variable Rate Application (VRA):** Optimize chemical and fertilizer use through zone-based variable rate control, maximizing yield while reducing environmental impact.
- **Multi-Section Control (Up to 7 Sections):** Manage up to seven independent sections with precision rate adjustments, enabling tailored application across complex field conditions.
- **Real-Time Data Processing:** Process sensor and location data on the go, allowing immediate adjustments for consistent performance and smarter field management.
- **User-Friendly Interface:** Simplify operations with an intuitive menu design that minimizes operator training time and enhances workflow efficiency.
- **Advanced Sensor Compatibility:** Integrate seamlessly with a wide range of agricultural sensors to expand functionality and improve decision-making accuracy.
- **Rugged & Reliable Hardware:** Built for tough field environments, Nova 200's robust electronics ensure dependable performance and long-term durability in all conditions.

3.2 Product Overview



Note: *The final product may differ from the description depending on the requirements and customer specifications.*

Sr. #	Part Name
1.	Menu Button
2.	ESC Button
3.	RATE Button
4.	Auto Section Control Button
5.	Auto Rate Control Button
6.	Main Valve Switch
7.	Section Valve Switches (7 individual section switches are available in NOVA 200)
8.	Rate Switch
9.	Navigation Buttons (Up, Down, Left, Right)
10.	Display Unit

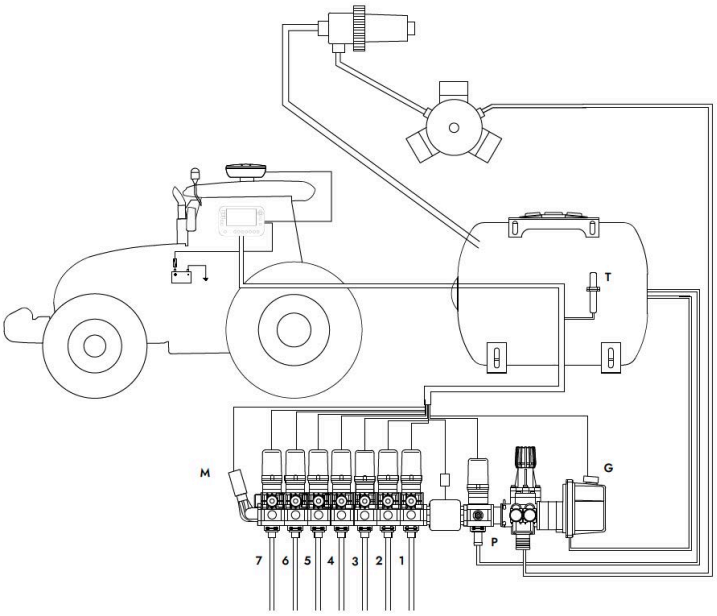


Specifications	Values
Dimensions	25.4 cm x 17 cm x 11.8cm
Weight	1.2 KG
Operating Temperature	-20°C Min. 70°C Max.
Display	20 × 4 DOT MATRIX Display
Power	12VDC or 24VDC (Depends on Setup)
Protection	IP 65

4. Installation

4.1 Device Installation

⚠	The installation must be carried out by a skilled professional only.



Typical Installed Setup on a Tractor

Step 1. *Unbox the equipment and place the NOVA 200 computer on a flat and clean surface.*

☞	- NOVA 200 can be installed in TWO Configurations. - Using the Triple Suction cup mount - Using the Screw mount

Step 2. *Choose your preferred installation configuration (triple suction mount or screw mount) and ensure that the mounting bracket is securely fastened.*



⚠	If the mounting bracket is not securely fastened, it may result in movement or detachment during operation and may cause damage to product or operator.

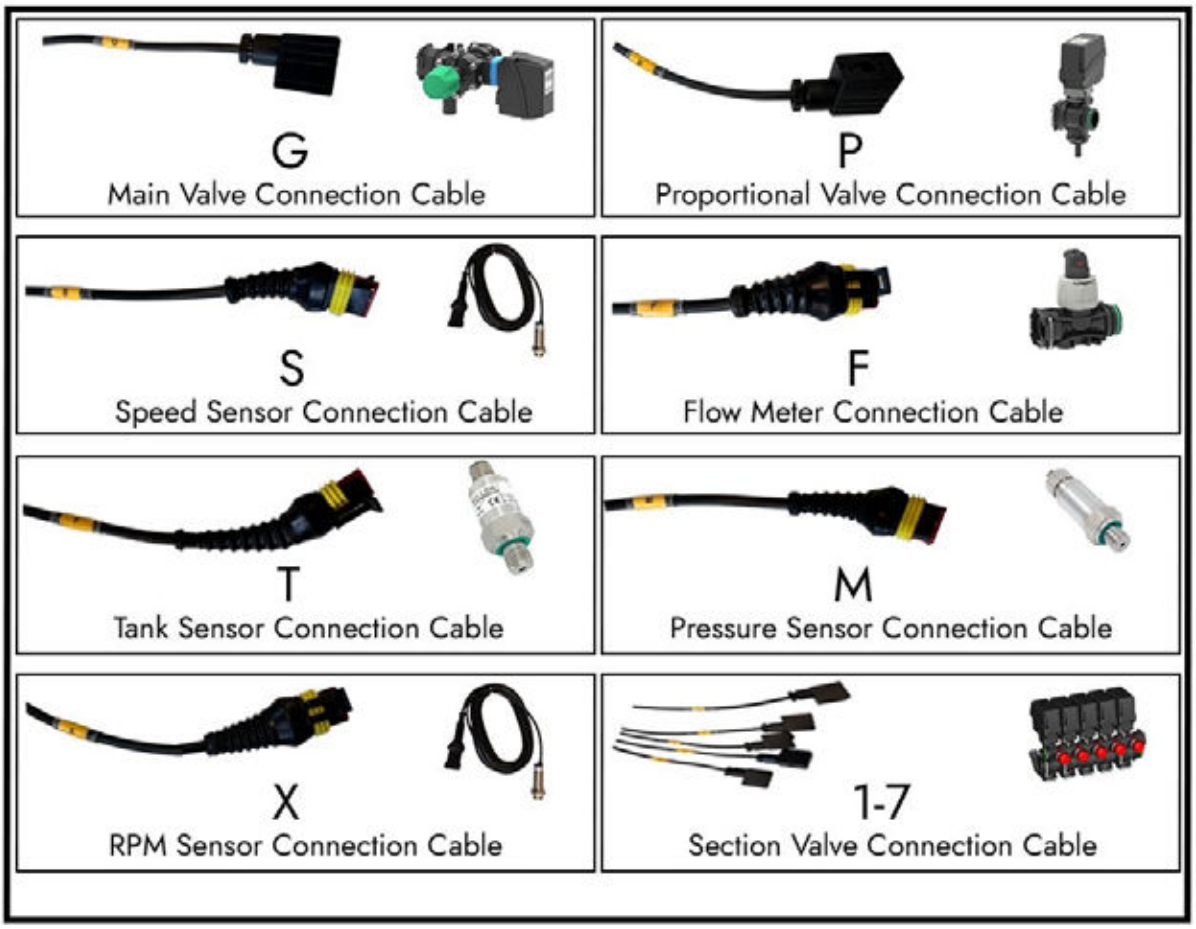
Step 3. *For proper connection of the Nova 100 Rate Controller to the tractor, the harness must be constructed to ensure reliable power supply, signal transmission, and compatibility with existing equipment.*

Step 4. *The harness includes a dedicated power line (preferably fused) connected directly to the tractor battery (12V or 24V, depending on system) as shown below.*

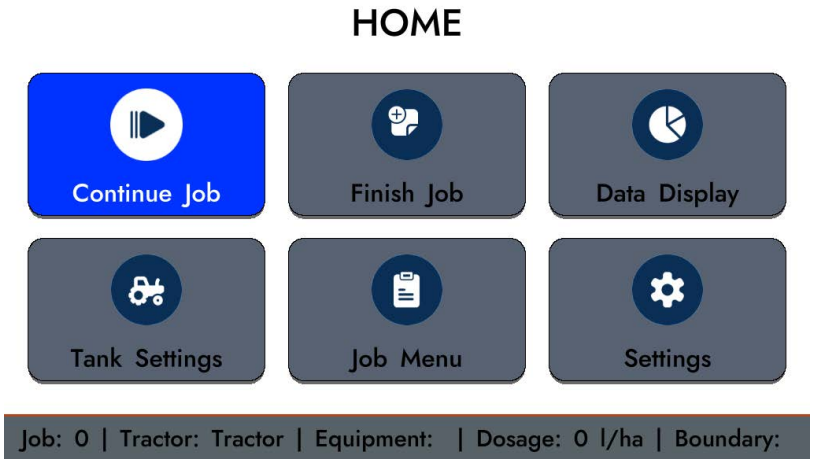


Connectors must be waterproof (IP65 or higher rated) for agricultural environments.

Step 5. *Clearly labeled wires for boom control, flowmeter, pressure sensor, speed input (e.g., GPS or wheel sensor), and section valves as shown below.*



5. Software Operations



The NOVA 200 offers a simplicity and easy to use Menu interface which is explained below.

5.1 Continue Job

The Continue Job function allows the operator to resume spraying operations from the exact point where the previous job was interrupted. This feature is especially useful when the spraying process is paused due to refilling, weather changes, or transport between fields.

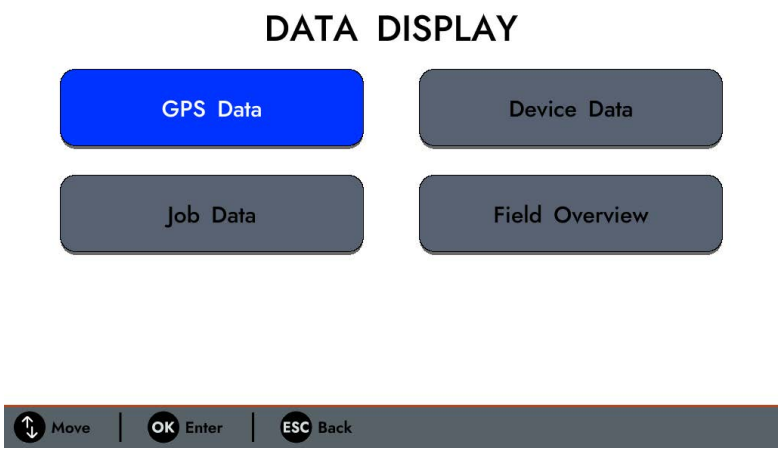
When a job is resumed, all previously recorded data such as sprayed area, applied quantity, and GPS location are reloaded. This ensures data integrity and prevents double application or gaps in coverage.

5.2 Start New Job

The Start New Job function allows the operator to begin a new spraying operation using the same configuration settings (such as desired spray rate, nozzle type, boom setup, and control mode) from the previous job, while resetting all job statistics to zero.

This feature is ideal when the same chemical mix or application parameters are used in a different field or section, ensuring consistent performance while maintaining clean job data records.

5.3 Data Display



5.3.1 GPS Data

The GPS Data screen displays real-time positional and navigation information, including Longitude, Latitude, Altitude, Number of Satellites, HDOP (Horizontal Dilution of Precision), Course (direction of movement), and Speed.

5.3.2 Job Data

The Job Data screen provides key application parameters such as Job ID, Applied Area, Calculated Area, Total Applied Volume (liters), Average Application Rate, Target Rate, and Selected Nozzle Type.

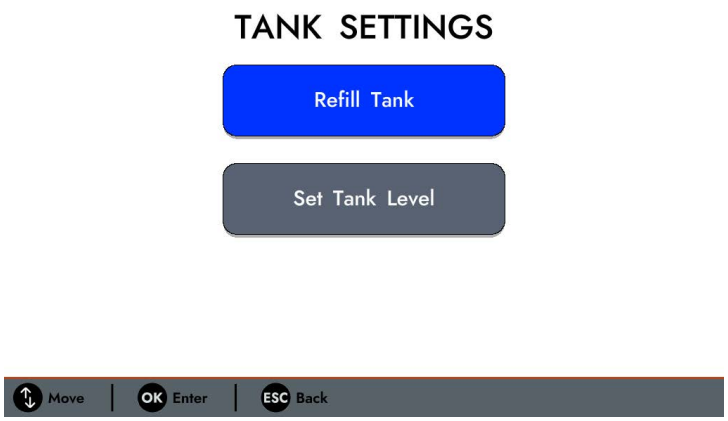
5.3.3 Device Data

The Device Data screen provides system diagnostics and operational statistics, including Operating Voltage, Software Version, CPU Temperature, Total Traversed Area, Total Sprayed Volume, Total Operation Time, and the device's current IP Address.

5.3.4 Field Overview

The Field Overview screen provides a summarized view of your current spraying task — showing real-time operational data, total application statistics, and progress within the selected field. It helps operators monitor efficiency and ensure correct application.

5.4 Tank Settings



The Tank Settings menu allows the user to manage the tank level data used during spraying operations.

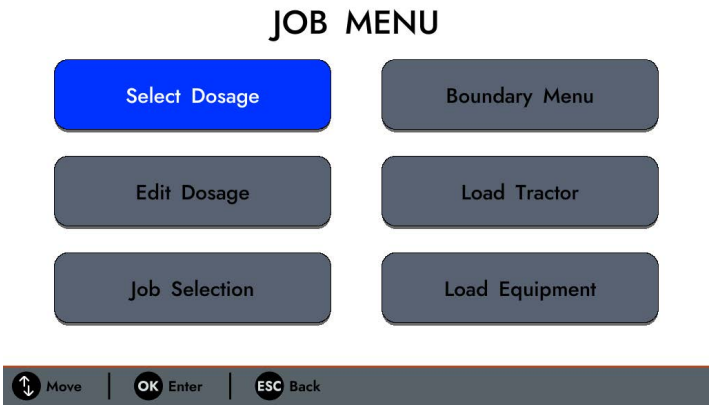
5.4.1 Refill Tank

This option instantly sets the tank level to its maximum capacity as defined during implement setup. For example, if your implement was configured with a 2500-liter tank, selecting “Refill Tank” will reset the tank level to 2500 L.Use this option after a full refill in the field to quickly reset the system.

5.4.2 Set Tank Level

This option allows you to manually enter the current liquid level in the tank. Useful if you only partially filled the tank.

5.5 Job Menu



5.5.1 Select Job

The Select Job function enables the operator to choose from up to nine (9) saved jobs, each with its own unique spray settings.

Every job profile can store individual parameters explained below:

5.5.1.1 Desired spray rate (L/ha)

Set the appropriate spray rate. The operator can later shift between different rates without manually entering the setting value every time.

5.5.1.2 Nozzle Type

This allows the operator to quickly switch between different application types without manually re-entering settings each time.

5.5.2 Edit Job

The Edit Job function allows the operator to modify the parameters of any previously saved job without creating a new one.

You can update values explained below:

5.5.3 Select Memory

The Select Memory function allows the operator to recall jobs saved in the Nova 200's internal memory.

This feature provides quick access to previously saved job data, making it easy to review or reuse existing spraying configurations.

Each saved job includes its specific parameters explained below:

5.5.3.1 Job Progress Data

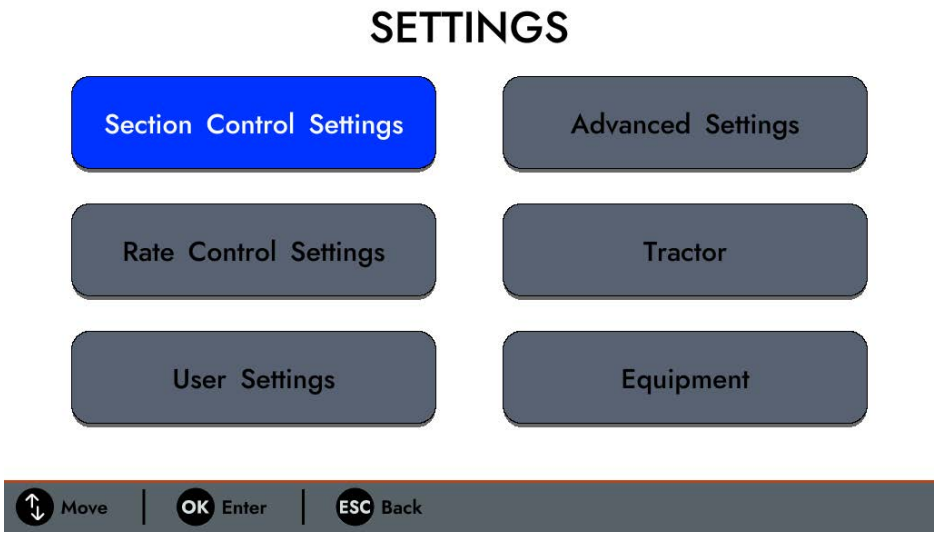
This allows the operator to quickly review the job progress data including Sprayed Quantity, Area and more.

5.5.4 Boundary Menu

The Boundary Menu allows the operator to recall previously saved field boundaries from the Nova 200's internal memory. This feature ensures that field limits are correctly identified for precise spraying, automatic section control, and accurate area calculation.

A boundary defines the outer limits of the field, helping the system recognize where spraying should start and stop.

5.6 Settings



The Settings Menu allows the operator to configure all system parameters of the Nova 200 to ensure accurate and efficient spraying performance.

Through this menu, users can adjust control preferences, system calibration, display units, and hardware options according to field requirements or operator preferences.

The Settings menu is divided into several categories for easy navigation.

5.6.1 Tractor

The Tractor Settings menu allows the operator to create, edit, and load tractor profiles used with the Nova 200 system.

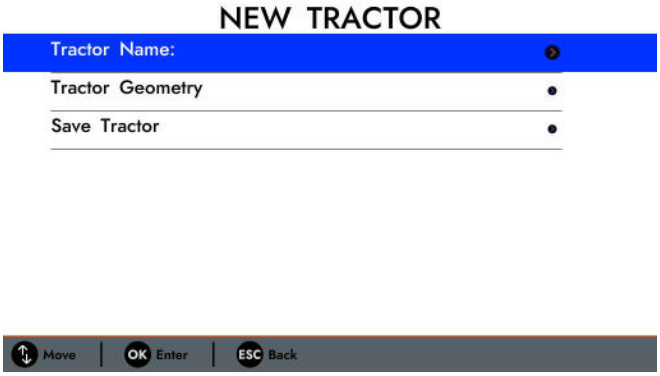


5.6.1.1 Load Tractor

The Load Tractor function allows the operator to recall a previously saved tractor profile from the Nova 200's internal memory.

5.6.1.2 Create Tractor

The 'Create Tractor' function allows users to define a new tractor profile by entering the tractor's geometric parameters.



Tractor Name: Allows the user to assign a custom name to the tractor profile for easy identification.

Tractor Geometry: Accurate tractor geometry is essential for precise guidance, tracking accuracy, and optimal alignment between the GPS antenna and the implement. This section enables the user to input the critical structural dimensions of the tractor to ensure correct system behavior during operation.

Wheelbase: This is the distance between the front and rear axles of the tractor. Accurate wheelbase entry is important for modeling the tractor's turning dynamics, particularly during curved path following and headland maneuvers.

Rear Axle Connection Point: This defines the distance between the rear axle of the tractor and the implement connection point. This setting is particularly important for accurately modeling the implement's trailing behavior during headland turns and reverse maneuvers.

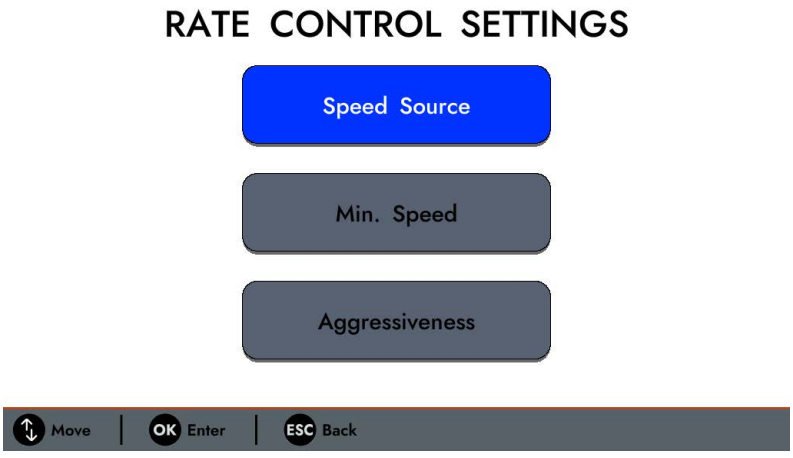
GPS Antenna Rear Axle: This defines the distance from the GPS antenna (typically mounted on the tractor roof) to the rear axle. Accurate configuration of this parameter ensures precise positioning and correct timing for section control operations.

5.6.1.3 Edit Tractor

Allows the user to modify an existing tractor profile by updating its name or geometric parameters such as wheelbase, implement offset, and antenna position. This ensures the stored configuration remains accurate and aligned with any physical changes made to the vehicle.

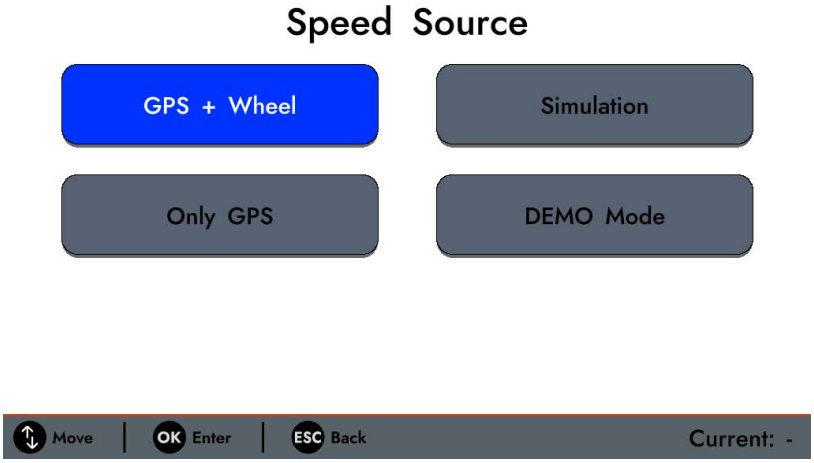
5.6.2 Rate Control Settings

The Rate Control Settings menu is used to configure how the system adjusts chemical application rates based on the machine's travel speed. These settings are essential for accurate and stable rate control in the field.



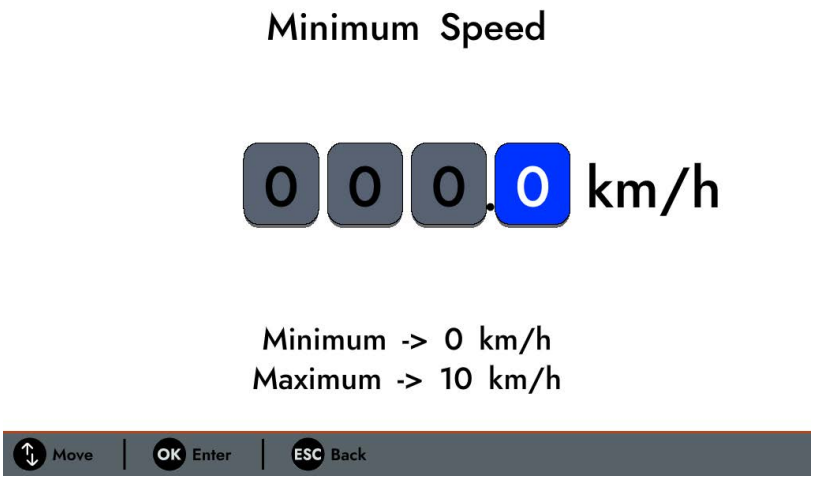
5.6.2.1 Speed Source

Wheel + GPS: Combines signals from both the wheel sensor and GPS receiver.
Only GPS: Uses only satellite-based positioning to estimate speed.
Only Wheel: Uses only the wheel sensor.
Simulation: Generates artificial speed for testing purposes.
DEMO Mode: Used for demonstration or training; not suitable for real application.



5.6.2.2 Min. Speed

This sets the minimum travel speed at which the system will attempt to control the flow rate. If the vehicle drops below this speed, the controller will stop and display an error.



5.6.2.3 Aggressiveness

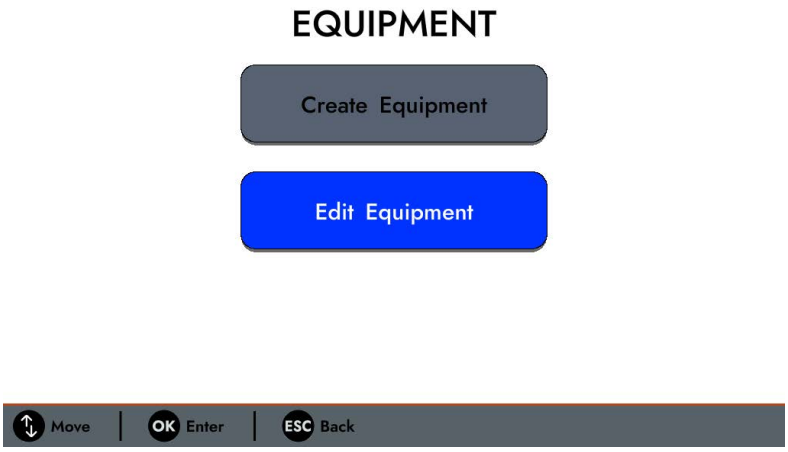
Defines how quickly the system reacts to speed changes and adjusts the flow rate accordingly.

- Higher aggressiveness: Faster response, ideal for uneven terrain or frequent speed variation
- Lower aggressiveness: Smoother changes, prevents overshooting or oscillations.



5.6.3 Implement

The Implement Settings menu allows the operator to create, edit, and load implement profiles used with the Nova 200 system. Each implement profile stores key configuration parameters such as boom width, number of sections, valve types, and sensor calibration. This enables fast setup when changing between different sprayers or implements without re-entering all data manually.

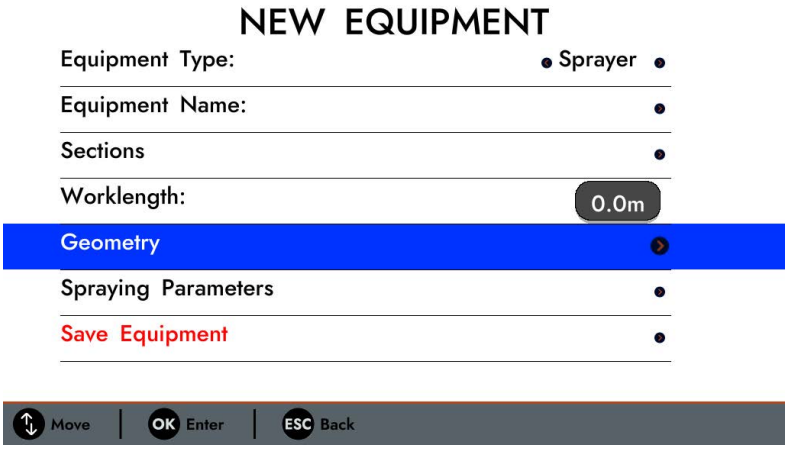


5.6.3.1 Load Implement

Allows the user to select and activate a previously saved implement profile. Loading an implement ensures that the system uses the correct parameters—such as working width, section configuration, and offsets—during field operations.

5.6.3.2 Create Implement

Allows the user to define a new implement profile—such as a sprayer or other equipment—by entering its key physical parameters and operational characteristics. This enables precise control and accurate application during field operations.



5.6.3.2.1 Implement Type

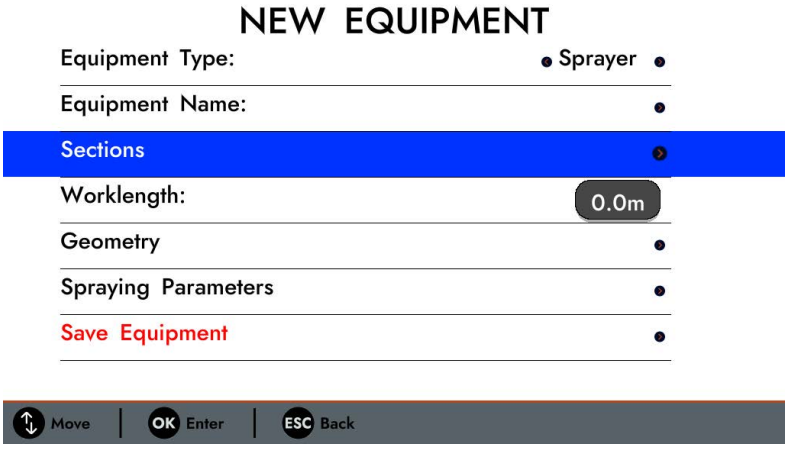
This option allows the user to select the type of implement being configured. Use the Right Arrow button to cycle through available types, such as Sprayer or Others.

5.6.3.2.2 Implement Name

Allows the user to assign a unique, easily identifiable name to the implement profile. Press the OK button to enter or modify the name.

5.6.3.2.3 Sections

On this screen, the user defines how the implement is divided into individual spray sections. Proper section configuration is essential for enabling Section Control, which automatically activates or deactivates sections to prevent overlapping and ensure application only in untreated areas.



5.6.3.2.3.1 Nozzle Spacing

Defines the distance between two adjacent nozzles on the boom. This value directly influences the accuracy of working width and section calculations, and must be configured according to the physical setup of the implement.

5.6.3.2.3.2 Section

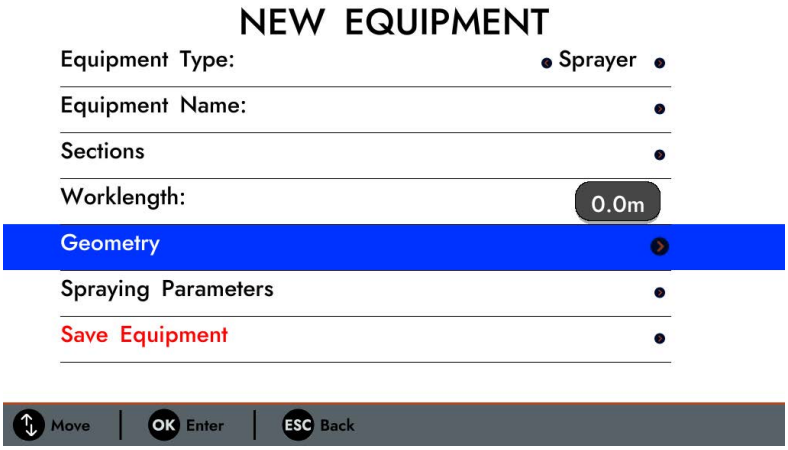
For each section, you enter the number of nozzles (e.g., 6). The system calculates the section width automatically by multiplying the number of nozzles by the configured nozzle spacing. Accurate nozzle count input ensures precise section width calculation and effective application control.

5.6.3.2.4 Worklength

The total working width of the implement is determined automatically by summing the individual widths of all defined sections. Each section width is calculated based on its nozzle count and the configured nozzle spacing.

5.6.3.2.5 Geometry

For implements defined as sprayers, this section allows the user to configure key geometric parameters. These include critical distance measurements such as the horizontal and vertical offsets between the implement and the tractor. Accurate geometry settings are essential for correct positioning, section control timing, and field coverage calculations.



5.6.3.2.5.1 Implement Type

Specifies whether the implement is Towed (pulled behind the tractor) or Carried (mounted directly on the tractor). This setting affects how the system calculates implement position and movement dynamics.

5.6.3.2.5.2 Application Point Wheel Axle Distance

This is the distance from the spray or working point (e.g., nozzle boom) to the center of the implement wheels. This parameter influences the timing of section ON/OFF commands by adjusting for the physical position of the application point relative to the GPS location.

5.6.3.2.5.3 Wheel Axle Connection Point Distance

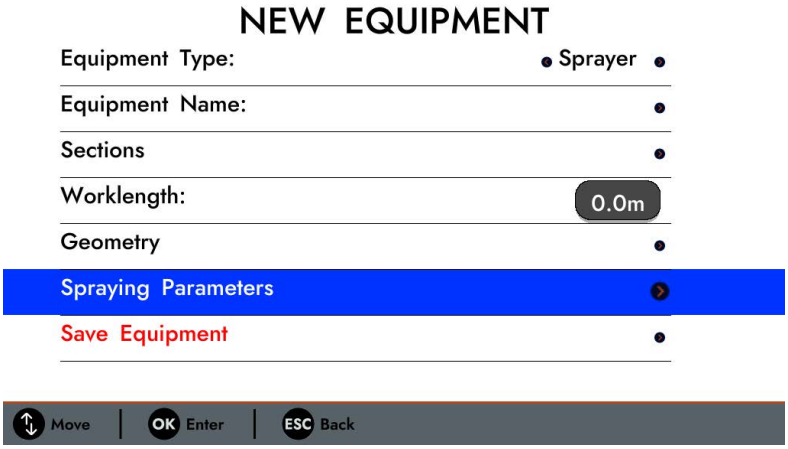
Defines the distance from the center of the implement's axle (wheels) to the connection point on the tractor (such as the drawbar or hitch). This parameter is important for accurate implement tracking and maneuvering behavior, especially during turns and reverse operations.

5.6.3.2.5.4 Application Point Connection Point Distance (Carried Implement)

This is the distance from the application point (e.g., nozzle boom) to the implement's connection point on the tractor. For carried sprayers, this value is critical for accurately timing section control events and ensuring correct GPS-based positioning of the application area.

5.6.3.2.6 Spraying Parameters

This screen allows the user to configure the core technical parameters that influence flow rate and application accuracy during spraying operations. Correct setup of these values is essential for achieving precise rate control and consistent field performance.



5.6.3.2.6.1 Valve Type

Select the type of control valve installed on your implement. Available options typically include 3-Way and 2-Way valves. Selecting the correct valve type is essential for proper flow regulation and system response.

5.6.3.2.6.2 Flowmeter Constant

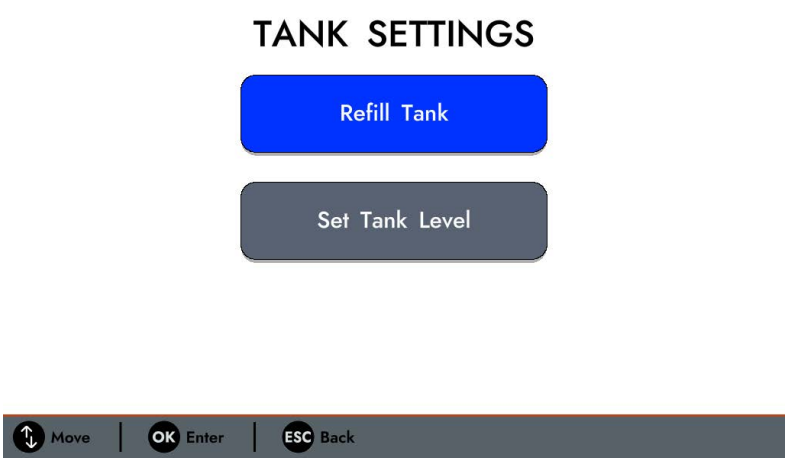
Defines the number of pulses per liter (pls/L) generated by the flowmeter. This value is provided by the flowmeter manufacturer and is essential for accurate measurement of the applied volume during operation.

5.6.3.2.6.3 Wheel Constant

Defines the number of centimeters traveled per pulse received from the wheel or speed sensor. This constant calibrates the distance measurement, enabling the system to accurately calculate the area covered. The value can be entered manually or determined automatically by driving a 100-meter distance with the tractor during calibration.

5.6.3.2.6.4 Tank Setup

In this menu, you can configure key tank-related parameters, including Tank Capacity, Minimum Tank Level, and the Tank Level Source. Proper configuration ensures accurate monitoring of liquid availability during spraying operations.



Tank Capacity
Defines the total volume of the spray tank in liters. This value is used by the system to calculate the remaining liquid, monitor consumption, and provide warnings when the tank level is low.

Tank Minimum Level
Sets the minimum liquid level (in liters) at which the system will trigger a low tank warning. This helps prevent unexpected interruptions during application by alerting the operator before the tank runs empty.

Tank Level Source
Specifies how the system determines the current tank level. Two options are available:

Manual: The operator manually enters the tank fill level after each refill.
Sensor: The system automatically reads the tank level from a connected liquid level sensor.

- Manual
- Automatic
- Max Tank Sensor

5.6.3.2.7 Pressure Sensor

The Pressure Sensor function defines how the Nova 200 measures or estimates system spray pressure. Correct pressure detection is essential for stable application rate control and nozzle performance.

Depending on the sprayer configuration, the operator can choose between three modes:

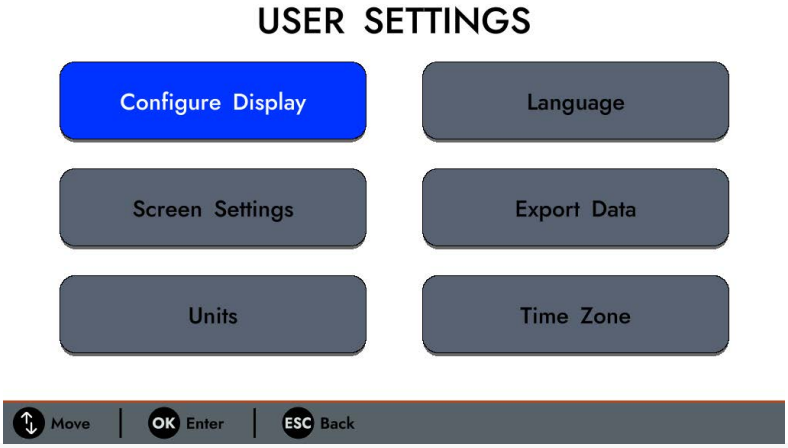
- Sensor
- Estimation
- None

5.6.3.3 Edit Implement

Enables the user to modify an existing implement profile by updating parameters such as implement type, working width, section count, and offset values. This ensures the implement configuration remains accurate for field operations.

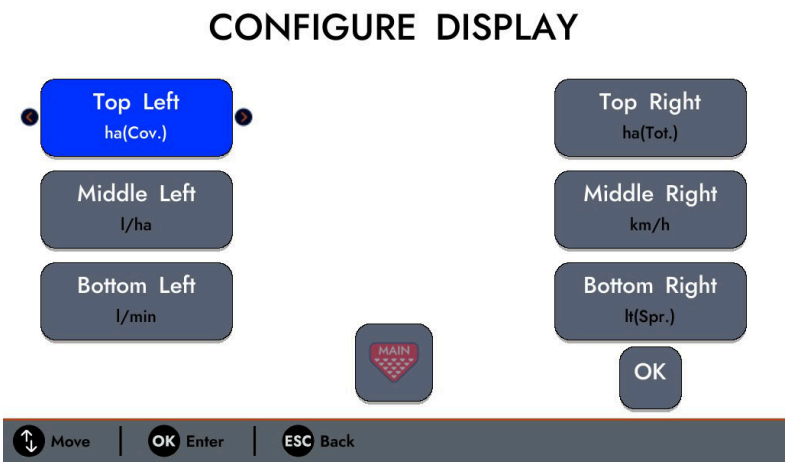
5.6.4 User Settings

The User Settings menu allows the operator to personalize the Nova 200 interface and display options. These settings do not affect spraying performance but improve ease of use, readability, and localization for different operators and environments.



5.6.4.1 Configure Display

The Configure Display menu lets the operator choose which information appears on the main screen during operation.



The screen has six zones: Top Left (TL), Top Right (TR), Middle Left (ML), Middle Right (MR), Bottom Left (BL), and Bottom Right (BR).

Following are the available Display Options:

Position	Display	Description
Top Left (TL)	ha (Tot)	Total sprayed area
Top Right (TR)	ha (Covered)	Active sprayed area
Middle Left (ML)	(l/ha)	Total Sprayed Volume Per Hector
Middle Right (MR)	(km/h)	Forward speed
Bottom Left (BL)	(l/min)	Current Flow Rate
Bottom Right (BR)	lt.(Spr.)	Remaining tank liquid

5.6.4.1.1 Language

The Language menu allows the user to select the preferred language for the NOVA 200 interface.



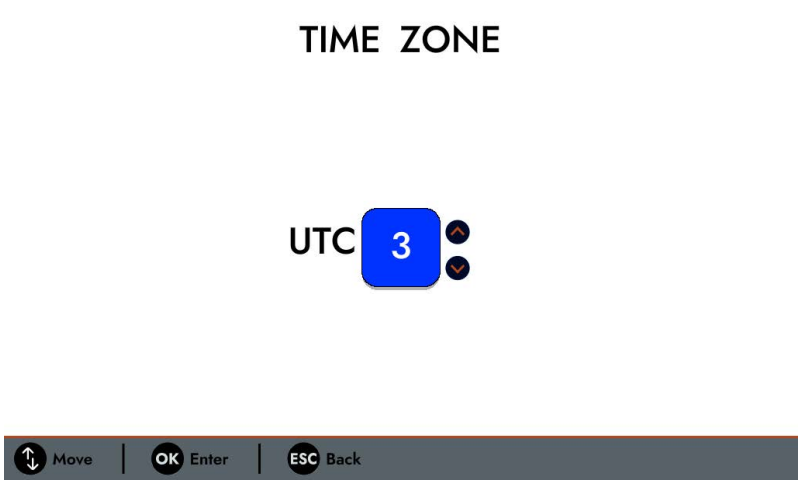
5.6.4.1.2 Export Data

The Export Data function allows the operator to save and transfer job information such as sprayed area, quantity, and boundaries to an external device.



5.6.4.1.3 Time Zone

The Time Zone option lets the operator set the local time and date used in all jobs and reports.

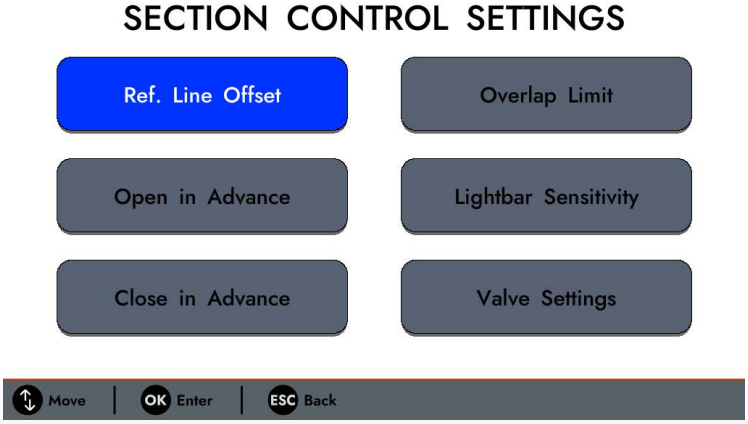


5.6.4.1.4 Units

The Units menu lets the operator select the preferred measurement system (Metric or US) for all displayed and recorded values.

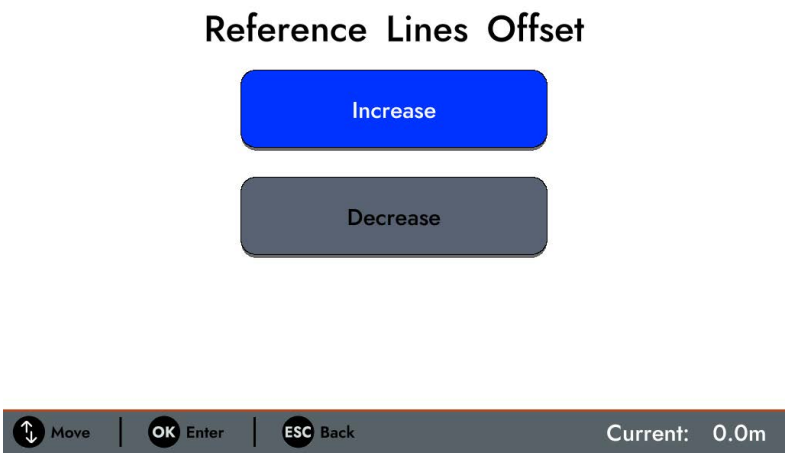
5.6.5 Section Control Settings

The Section Control Settings menu defines how boom sections open and close during spraying for precise coverage and minimal overlap.



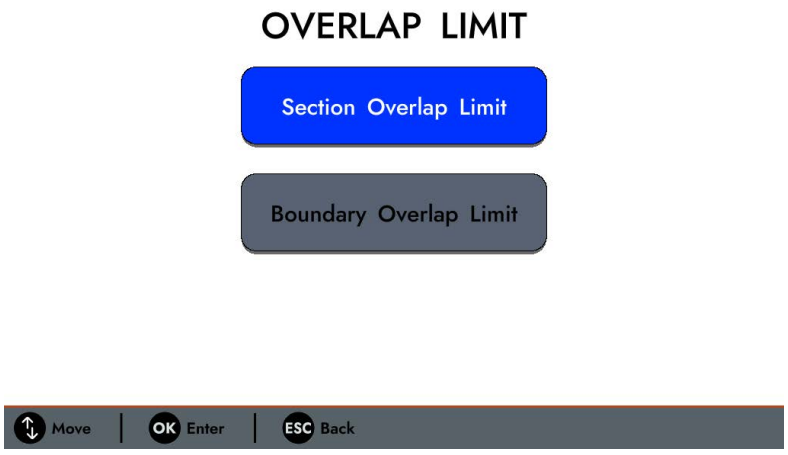
5.6.5.1 Ref.Line Offset

Adjusts the offset distance between the vehicle centerline and the actual boom position (e.g., GNSS antenna to boom). Use positive or negative values depending on implement mounting.

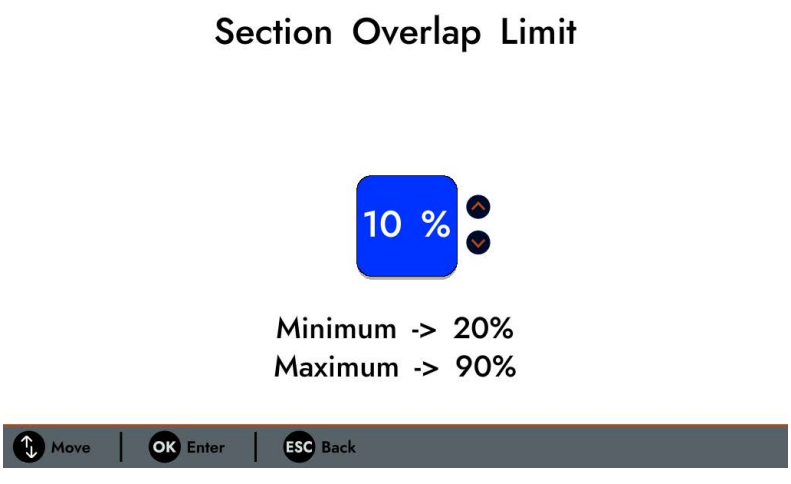


5.6.5.2 Section Overlap Limit

Sets the maximum allowed overlap area before sections automatically close.

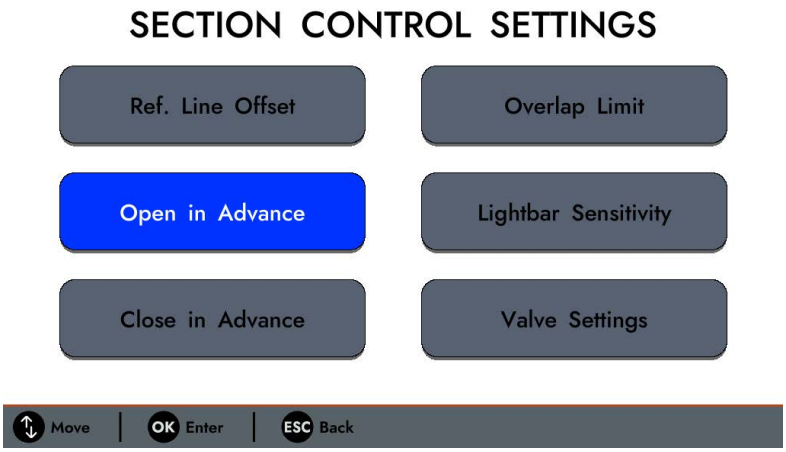


You can set Section or Boundry Overlap Limits by selecting any value.



5.6.5.3 Open in Advance

Opens sections slightly before reaching the untreated area to ensure full coverage.



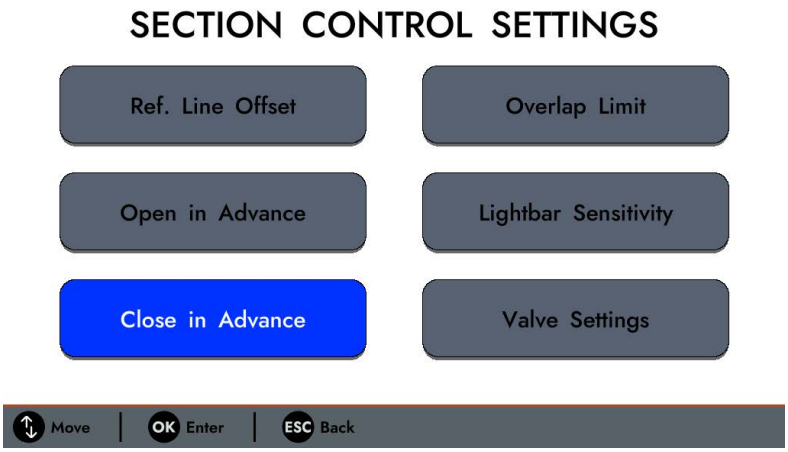
5.6.5.4 Lightbar Sensitivity

Adjusts steering guidance sensitivity in the lightbar display.



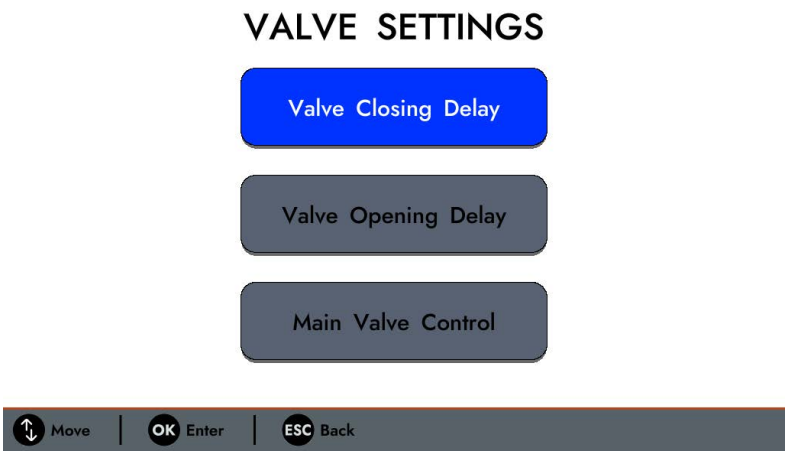
5.6.5.5 Close in Advance

Closes sections slightly before exiting the treated area to prevent double spraying.



5.6.5.6 Valve Settings

The Valve Settings menu adjusts how the main and section valves respond during spraying. The valve settings offers the following options.



5.6.5.6.1 Valve Opening Delay:

Sets the time (0–3 s) before a valve fully opens after activation.

> Use longer delay for slower valves

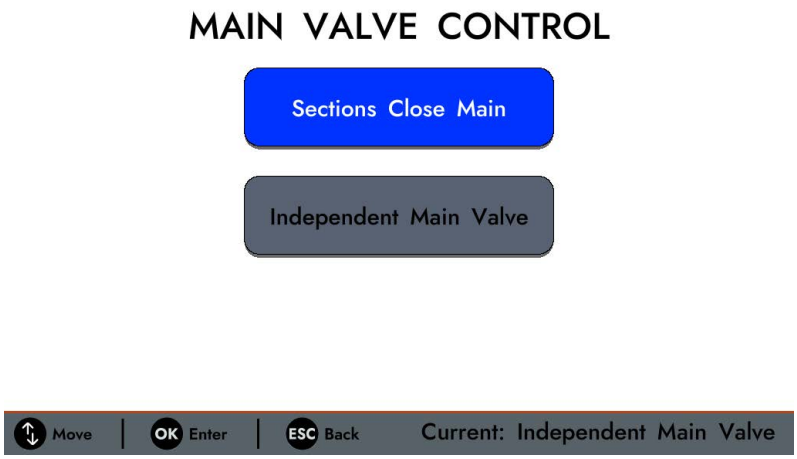
5.6.5.6.2 Valve Closing Delay

Sets the time (0–3 s) before a valve fully closes after deactivation.

> Helps prevent pressure drop or uneven coverage.

5.6.5.6.3 Main Valve Control

Defines how the main supply valve operates in coordination with section valves.



5.6.5.6.3.1 Sections Close Main

The main valve automatically closes when all sections are closed and reopens when any section opens.

> Recommended for automatic section control systems.

5.6.5.6.3.2 Independent Main Valve
The main valve is controlled manually and separately from section valves.

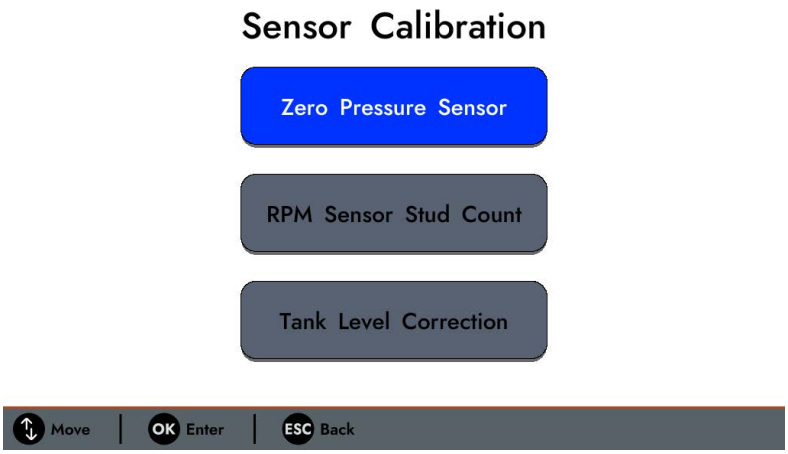
☞	Useful for testing or manual spraying modes.

5.6.6 Advanced Settings
The Advanced Settings menu provides access to system-level tools for calibration, diagnostics, connectivity, and maintenance of the Nova 200.

☞	These options are intended for experienced users or authorized service technicians.

5.6.6.1 Sensor Calibration
The Sensor Calibration menu is used to ensure all connected sensors provide accurate and reliable readings.

Calibration should be performed during setup or after sensor replacement.



5.6.6.1.1 Zeroing Pressure Sensor
Sets the pressure sensor's zero point when the system is not pressurized.
- Ensure there is no pressure in the spray line.
- Go to Advanced Settings Sensor Calibration Zero Pressure.
- Press OK to set zero reference.

☞	Perform this procedure only when system pressure is 0 bar.

5.6.6.1.2 RPM Sensor (Stud Count)
Defines the number of metal studs or magnets passing the speed sensor per wheel revolution.
- Count the studs or magnets on the wheel/sensor target.
- Enter the correct number under RPM Sensor Stud Count.
- Confirm with OK.

5.6.6.1.3 Tank Level Correction

Calibrates the tank level sensor for accurate volume indication.

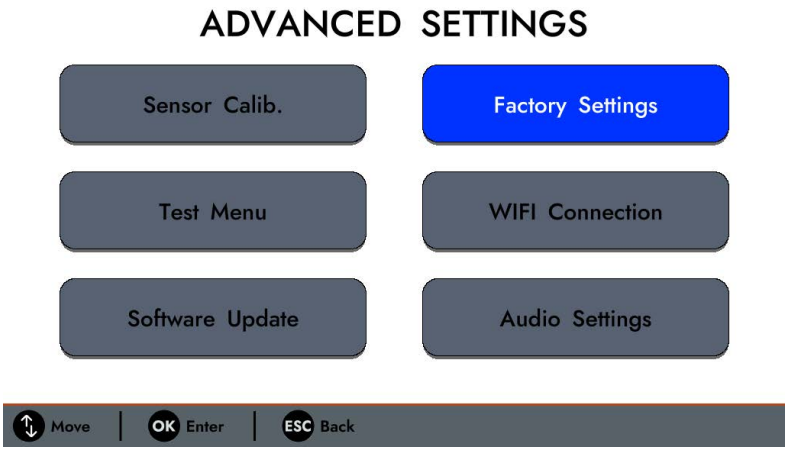
- Fill the tank step-by-step (e.g., 25%, 50%, 75%, 100%).
- Compare the displayed tank volume with the actual measured volume.
- Enter a correction factor (for example 1.0 = accurate, 1.1 = reads low, 0.9 = reads high).
- Press OK to save.

Correction factors can be entered as decimal numbers such as 1.0, 1.2, 1.5 depending on calibration results.

5.6.6.2 Factory Settings

The Factory Settings menu restores system parameters to their original default values.

Use these options carefully — some will erase stored data and custom settings.



5.6.6.2.1 Field Reset

Clears all job data, boundaries, and coverage maps but keeps device configurations and calibrations.

Use before starting a new season or when changing fields.

5.6.6.2.2 Factory Reset

Restores all system settings to original factory defaults, deleting jobs, calibrations, tractors, and implements.

Use only when required or instructed by service personnel.

5.6.6.2.3 Screen Default

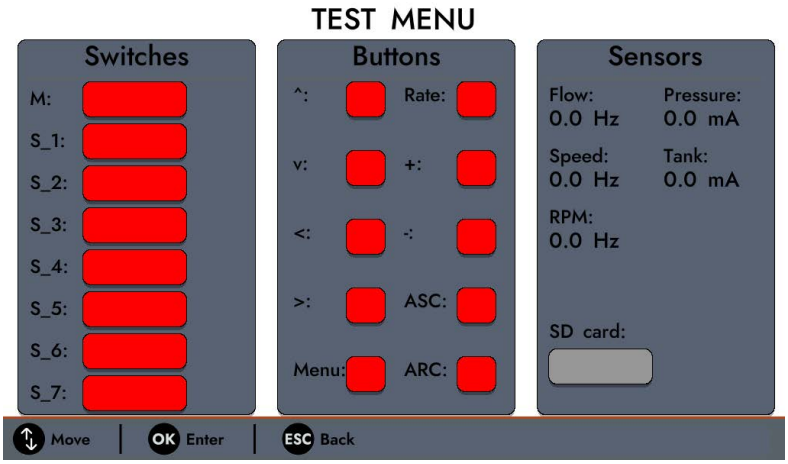
Resets the main display layout to the original factory configuration.

Useful if the display arrangement has been changed.

5.6.6.3 Test Menu

The Test Menu is a built-in diagnostic screen used to verify that the hardware components connected to the NOVA 200 terminal are functioning correctly.

It is particularly useful during installation, troubleshooting, or maintenance.



5.6.6.3.1 Switches Panel (Left Column)

This section tests the status of section control switches:

- M: Master switch
- S_1 to S_7: Individual section switches.

☞	Each switch turns green when it is detected as active or ON.

5.6.6.3.2 Buttons Panel (Middle Column)

This section monitors the physical buttons on the NOVA 200 device:

- Up / Down / Left / Right: Directional navigation buttons
- Menu / Rate / + / -: Function keys

☞	- Buttons that light up green are functioning and registering a press. - Red indicates no signal or no press detected at the time.

5.6.6.3.3 Sensors Panel (Right Column)

This section displays real-time values from connected sensors.

- **Flow:** Measured in Hz, shows pulses from the flowmeter
- **Speed:** Measured in Hz, shows wheel sensor activity
- **Pressure:** Displays analog value in milliamps (mA)
- **Tank:** Tank level sensor input in mA
- **RPM:** Shaft or engine speed in Hz
- **SD Card:** Indicates whether the SD card is inserted and functional

5.6.6.4 WIFI Connection

The WiFi Connection menu allows the operator to view and connect to available wireless networks directly from the NOVA 200 terminal.



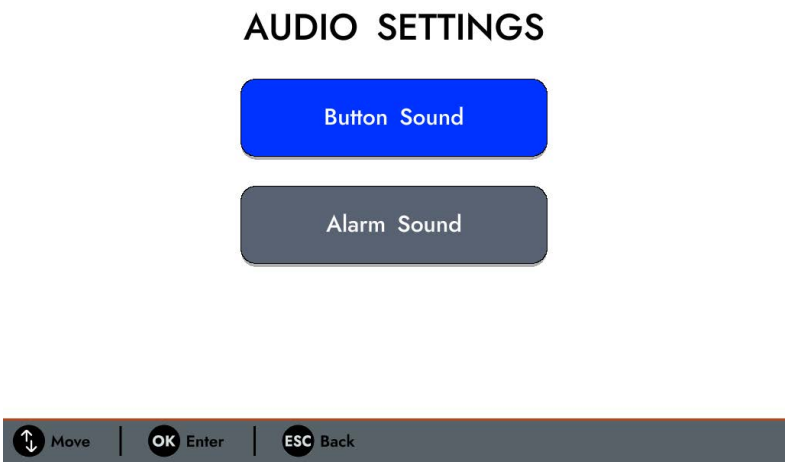
5.6.6.5 Software Update

The Software Update menu allows updating the Nova 200 firmware to the latest version for improved performance and new features.



5.6.6.6 Audio Settings

The Audio Settings menu lets the operator adjust system sounds such as Button Sound and Alarm Sound for improved usability in different environments.



6. Operating Instructions

The Operating Instructions section explains how to use the Nova 200 during spraying operations.

It covers first operation, new job instructions and operating features like boundary setup, section and rate automation, navigation guidance, pausing, and refilling procedures.

6.1 First Operation

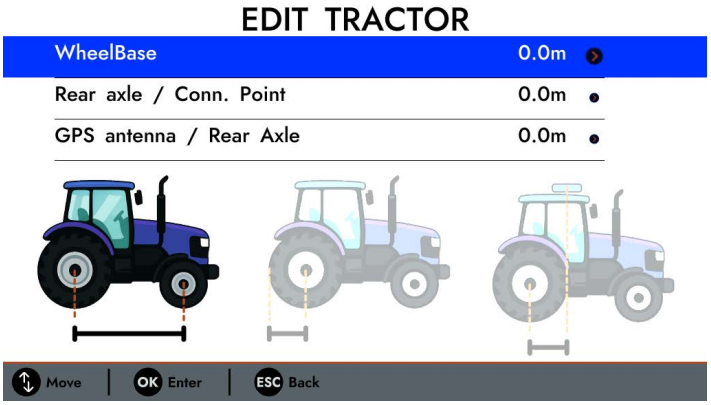
6.1.1 Power ON

- Switch ON the NOVA 200..
- Wait until the HOME MENU is displayed.

6.1.2 Tractor Geometry Settings

- Go to HOME → SETTINGS → Tractor.
- Tractor geometry values must be measured accurately with a tape measure
- The GPS antenna must be mounted exactly at the center of the tractor.

(⚠ Incorrect geometry or GPS position will cause guidance and section control errors.)

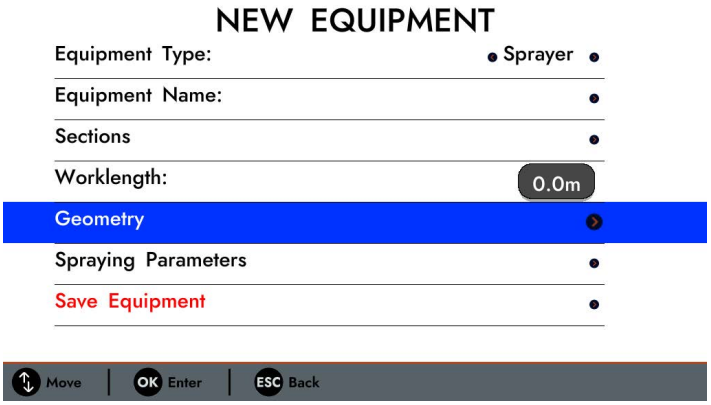


6.1.3 Implement Geometry & Configuration

ł Wō Ụo [wqD → SETTINGS → Implement.

- Implement geometry values must be measured accurately and entered correctly.
- Sprayer sections and parameters must be entered correctly.
- Typical flowmeter pulse values:
    - Arag Wolf:~250 pls/min
    - Arag Orion 2:~300 pls/min

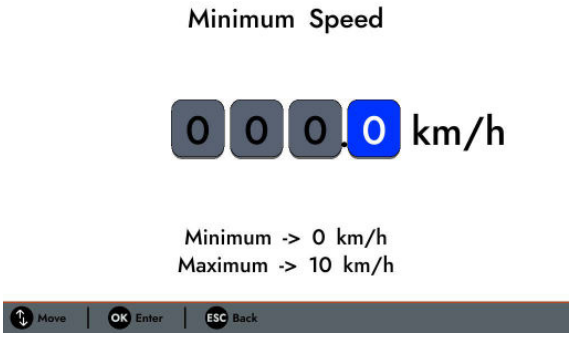
(⚠ Always check the flowmeter label and enter the exact pulse value written on it.)



6.1.4 Rate Control Settings

- Go to HOME → SETTINGS → Rate Control.

- Minimum Speed
    - Defines the minimum tractor speed at which spraying is allowed to start.
    - Below this speed, spraying will not start.
    - Spraying starts automatically once speed exceeds this value.



- Aggressiveness (Proportional Valve)
    - Defines how fast the proportional valve moves up and down.
    - Controls reaction speed to reach target rate.
      (⚠ Very high aggressiveness may cause pressure oscillation.)



6.1.5 Electrovalve Tests

- Electrovalves must always be tested before starting the Nova 200 sprayer. This ensures correct flow control, prevents leaks, and avoids uneven application.

6.1.5.1 Electrical (Sound) Test

1. Stop the tractor engine.
2. On NOVA 200, select Start New Job.
3. Ensure ARC is OFF.
4. Valve cables connected.
5. Activate section valves 1–2–3–4–5 (and 6–7 if available) and check sound.
6. Press + / – buttons and check sound from ARAG proportional valve (grey valve).

6.1.5.2 Liquid System Test

1. Start tractor.
2. Continue the same job.
3. Ensure ARC is OFF.
4. Open all section valves.
5. Open main valve.
6. Engage PTO.
7. Confirm spraying starts.
    a. Pressure Check
        i. If pressure is low, press + for ~10 seconds.
        ii. Pressure must be minimum 5 bar.

    b. Section Check
        i. Open/close sections one by one.
        ii. Check 3–5 bar liquid flow.
        iii. Section 1 = far left, last section = far right.
        iv. Red bypass calibration is done below 5 bar.
        v. Pressure gauge must always return to 5 bar when sections open/close.

6.1.6 Valve ON-OFF Delay Measurement

(ARC OFF, PTO ON)

- Close main valve → measure time until pressure reaches 0 bar → Enter as Valve Closing Time (typically 0.3 s)
- Open main valve → measure time until pressure reaches 5 bar → Enter as Valve Opening Time (typically 1.5 s)

6.1.7 Aggressiveness Fine Adjustment

1. Go to HOME → SETTINGS → Rate Control.
2. Typical starting value: 35%

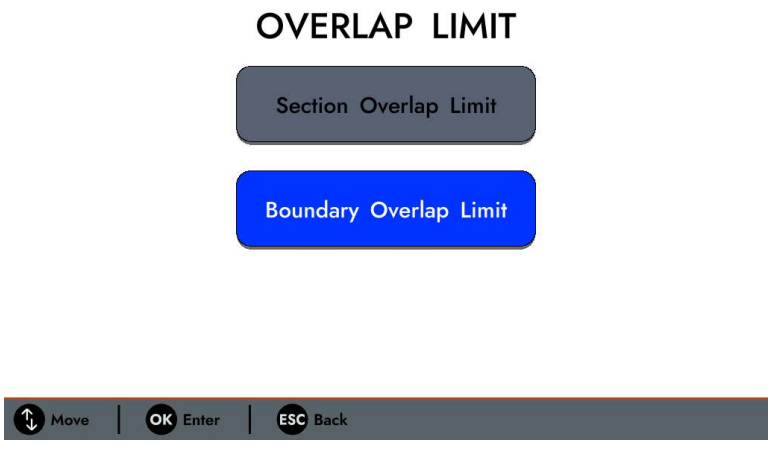
Procedure

a. Enable Simulation Mode
b. Turn ARC ON
c. Enter application rate (e.g. 200 l/ha)
d. Change simulated speed (e.g. 4–12 km/h)
(⚠ If pressure gauge oscillates at any speed, reduce aggressiveness. Final check should be done in the field with real working speeds.)



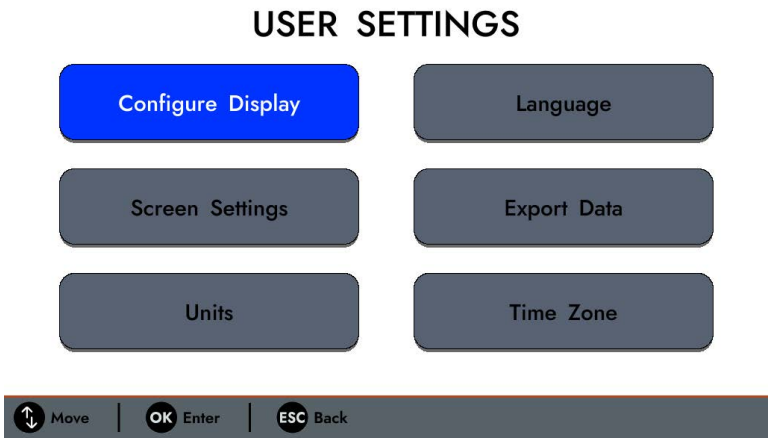
6.1.8 Section Control Parameters

1. Go to HOME → SETTINGS → Section Control.
2. Overlap Entry Distance (IN)
    a. Typical: 2 m
    b. 0 m → gaps (disease risk)
    c. 4 m → excessive overlap (chemical waste, burn risk)
3. Overlap Exit Distance (OUT)
    a. Typical: 1 m
4. Overlap Percentage
    a. Typical: 50%
    b. Common range: 30–70%



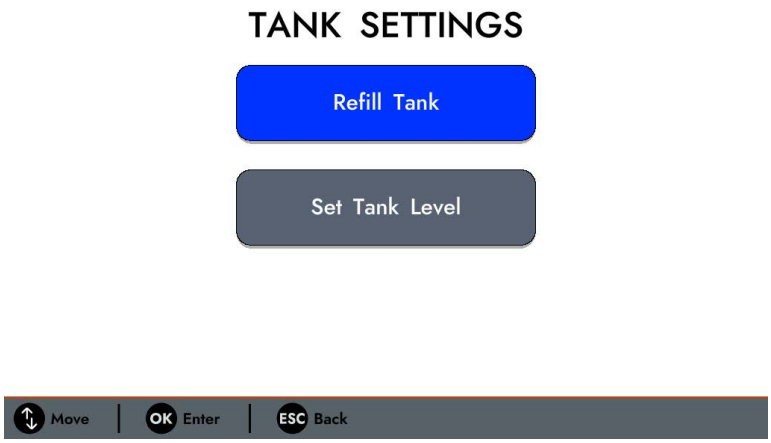
6.1.9 Screen Configuration – Tank Level

1. Go to HOME → SETTINGS → USER SETTINGS→ CONFIGURE DISPLAY
2. Add Tank Level to the top-right corner of the screen.
3. Save settings.



6.1.10 Tank Settings

1. Go to HOME → SETTINGS → TANK SETTINGS
2. Set Current Tank Level
3. If tank is refilled, select Tank Full.
   (⚠ Correct tank level is required for accurate job monitoring.)



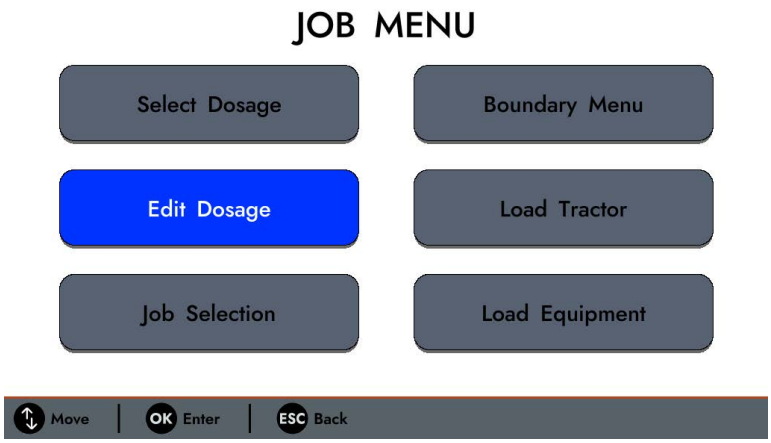
6.1.11 Select Tractor & Implement

1. Go to HOME → SETTINGS → TRACTOR or EQUIPMNET
2. Select correct Tractor
3. Select correct Implement



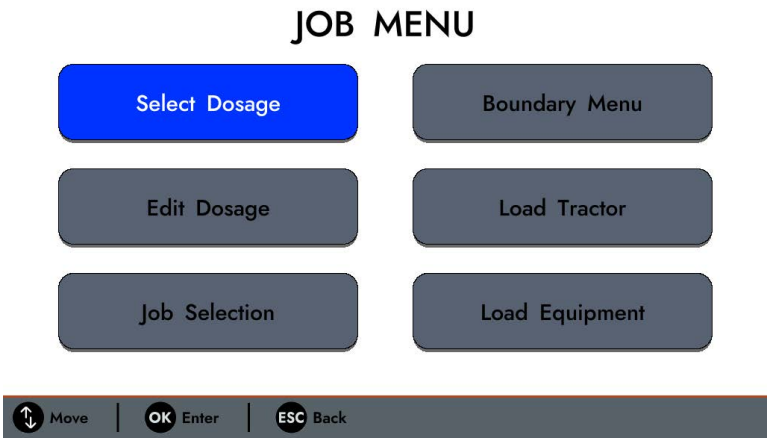
6.1.12 Select Application Rate

1. Go to HOME → JOB MENU → EDIT DOSAGE
2. Select the RATE for the desired application.
3. Confirm correct unit (l/ha, kg/ha).



6.1.13 Start New Job

1. Go to HOME → START NEW JOB
2. Confirm
    a. Tractor
    b. Implement
    c. Selected RATE
    d. Tank level
    e. Section control settings
3. Start Operation



6.1.14 First Operation – Sprayer Critical Check (Quick List)

1. Suction hose tight, no air leaks
2. Suction filter clean
3. Pump builds pressure smoothly
4. All line, section and nozzle filters clean
5. Ball valves open
6. Flowmeter installed in correct direction
7. Proportional valve moves freely
8. Section valves open/close correctly
9. Nozzles correct type and clean
10. Spraying starts only above minimum speed
11. Pressure and flow stable

Final Important Note
NOVA 200 controls rate and sections electronically, but accurate spraying depends on correct geometry, clean hydraulic system, correct valve behavior, and proper calibration.

6.2 Operating Features

6.2.1 Boundary Creation

To create a new field boundary, follow these steps.

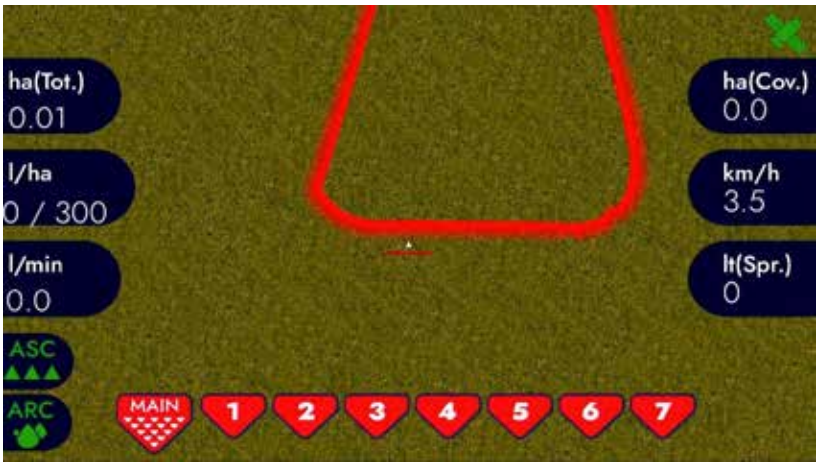
- Go to the Main Screen and Press the RIGHT ARROW.
- Tap the Boundary Icon on the screen.
- Select Left or Right to define which side of the tractor the boundary will be recorded.
- Drive around the edge of the field to trace the boundary line.

> Maintain steady speed and GPS signal while driving for accurate boundary shape.

6.2.2 Boundary Selection

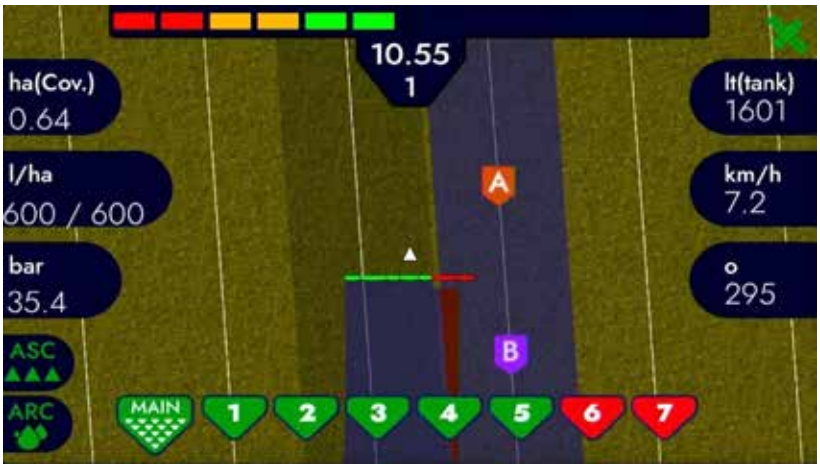
To use a previously saved boundary.

- Go to the Job Menu.
- Open the Boundary Menu.
- Select Boundary from the list of saved fields.
- Press OK to load the selected boundary.



6.2.3 Automatic Section Control – ASC

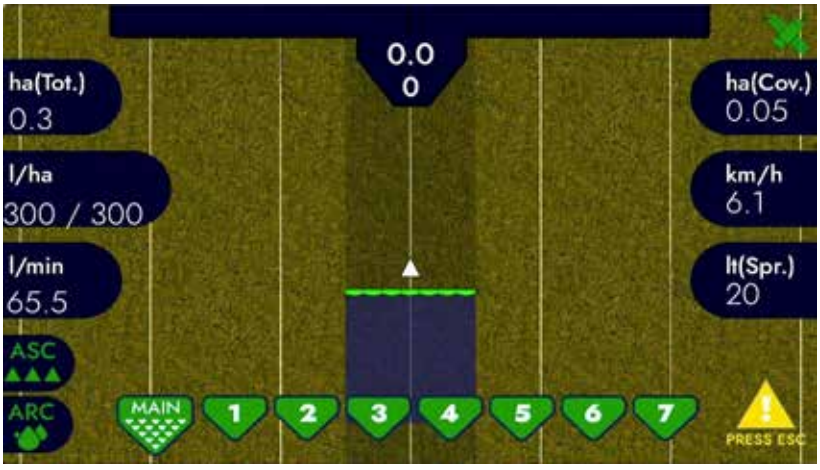
This feature is used to inhibit spraying of the previously sprayed areas. In order to activate this feature, GPS must be connected to the NOVA 200.



Pressing ASC button on the device will activate/deactivate this feature. If the light next to ASC button is on AND ASC symbol on the spraying screen is green means that automatic section control is ACTIVE.

6.2.4 Automatic Rate Control – ARC

This feature is used to control the spraying dosage independent of speed of the implement which has been attached to NOVA200 system.



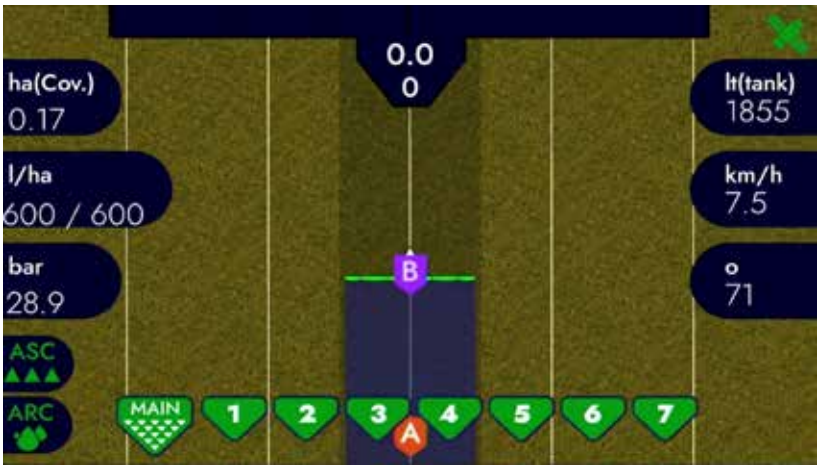
In order to use this feature on NOVA200, section widths and spraying parameters on the used implement must be entered in the implements menu. If the light is on next to ARC button and ARC symbol is green on the spraying screen means that this feature is ACTIVATED.

6.2.5 A – B Line

The A–B Line feature creates a straight guidance line for parallel driving and accurate spraying.
To create a straight guidance line for parallel driving.

- Go to the Main Screen and Press the RIGHT ARROW.
- Tap the A–B Line Icon on the screen.
- Drive forward at least 30 meters in a straight line.
- Press Mark Point A to set the starting point.
- After driving the distance, press Mark Point B to set the end point.
- The system will automatically create and activate the A–B guidance line.

> The longer the distance between A and B points, the more accurate the guidance line will be.



6.2.5.1 A – B Alignment

To adjust or shift the active A–B line position.

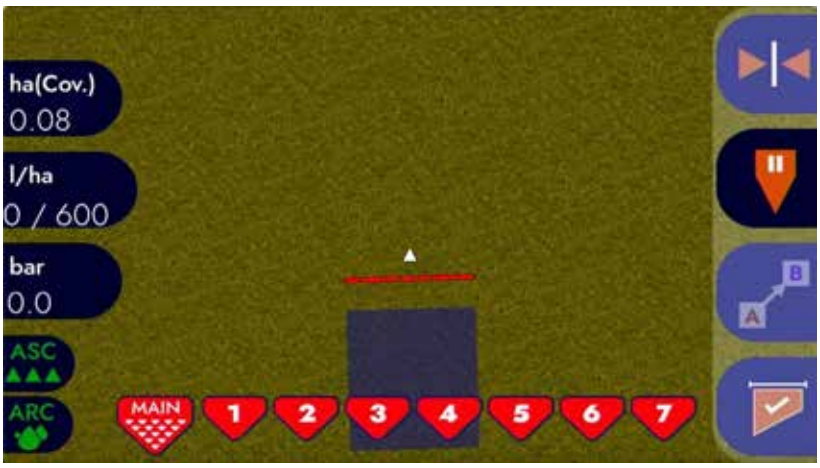
- Go to the Main Screen and Press the RIGHT ARROW.
- Tap the Line Shift Icon on the screen.
- Use the Left or Right controls to shift the guidance line to the desired position.
- Press OK to confirm.

> The longer the distance between A and B points, the more accurate the guidance line will be.



6.2.6 Pause Point

The Field Overview screen provides a summarized view of your current spraying task — showing real-time operational data, total application statistics, and progress within the selected field. It helps operators monitor efficiency and ensure correct application.



6.2.7 Tank Fill – Refill

If the tank becomes empty during operation, use these steps to restart normal operations.

- Refill the tank.
- On the Nova 200, go to Menu Tank Settings Refill Tank.
- Press OK to confirm.

☞	Always update the tank level on Nova 200 after refilling to keep spray rate and job data accurate.

7. Package Contents

7.1 Package Contents

Sr. #	Item Code	Description
1	TNOV200.01	Nova 200 Computer  Power Cable 2m 

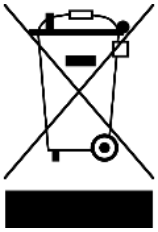
7.2 To be Ordered Contents

Sr. #	Item Code	Description
1	KYP030	Speed Sensor with 2m Cable 
2	SIM100	6m Valve Cable 
3	KYP036	Smart GNSS 

7.3 Optional Package Contents

Sr. #	Item Code	Description
1	SIM016	12 m Valve Cable 
2	KYP021	Level Sensor 0,4 Bar with 2m Cable 
3	KYP022	Pressure Sensor 25 Bar- G1/4 
4	KYP029KIT	Triple Suction Cup Mount 
5	KYP37	Screw Mount 
6	SAL400504	Kit Blister 3 Hole Plug Tractor Side 

8. Disposal of Used Packaging Material

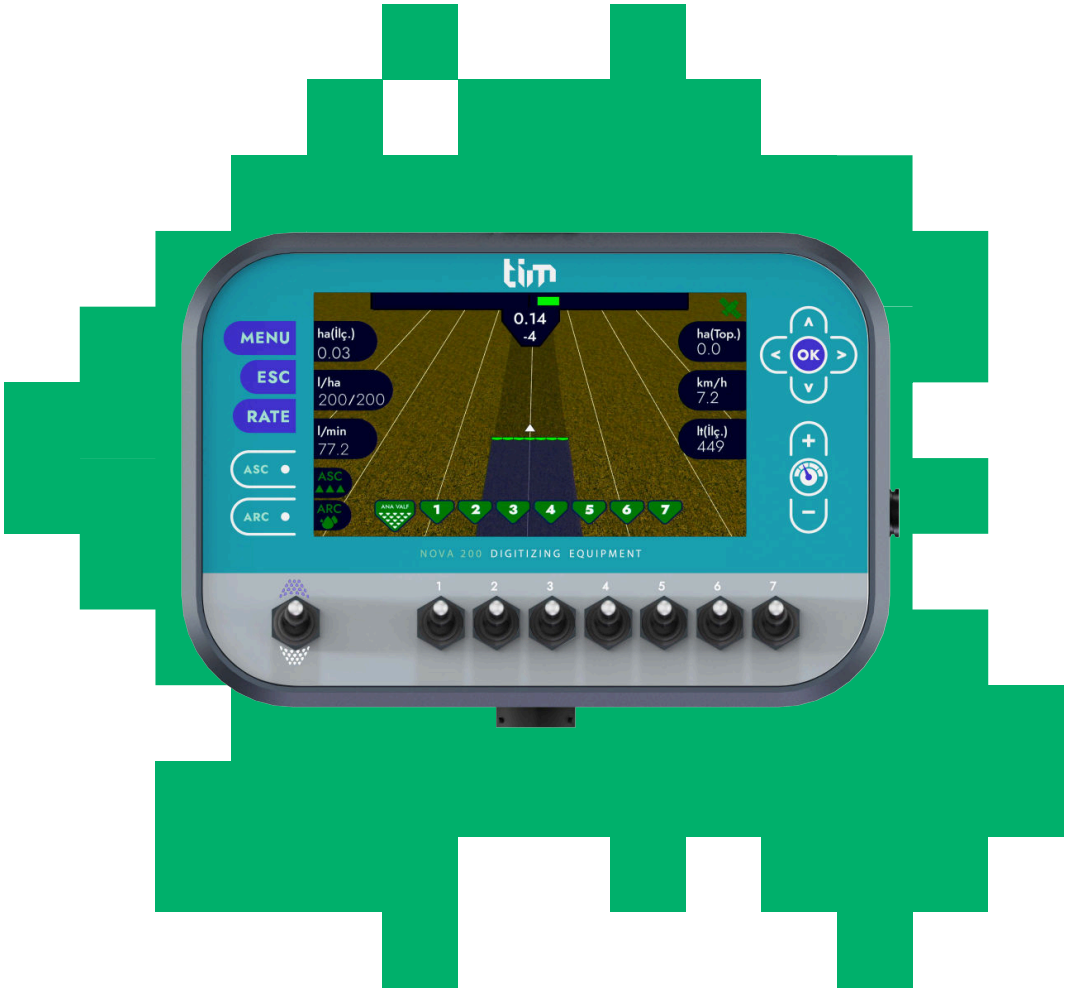


The meaning of the symbol on the product, its accessory or packaging indicates that this product shall not be treated as household waste. Please, dispose of this product at your applicable collection point for the recycling of electrical & electronic equipment waste. Alternatively, in some states of the European Union or other European states you may return your products to your local retailer when buying an equivalent new product. The correct disposal of this product will help save valuable natural resources and help in preventing the potential negative impact on the environment and human health, which could be caused as a result of improper liquidation of waste. Please ask your local authorities or the nearest waste collection center for further details. The improper disposal of this type of waste may fall subject to national regulations for fines.

8.1 For business entities in the US

If you wish to dispose of an electrical or electronic device, request the necessary information from your seller or supplier.

Changes to the text, design and technical specifications may occur without prior notice and we reserve the right to make these changes.



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